Bryan Wahyu Krishnaputra

Senior Backend Engineer — Go, Distributed Systems & Security

Depok, West Java, Indonesia · bryanwahyukp95@gmail.com · +62 899-0076-060 · linkedin.com/in/bryan-wahyu-krishnaputra · github.com/bryanwahyu

SUMMARY

Backend engineer with 12+ years of experience, starting in game backends in 2014 and now focused on distributed systems in Go. Scaled a social platform backend to 500K+ active users at <200ms p95 response time. Security engineering background: SOC tooling, an in-house SIEM for a regional government, and a DevSecOps scanning platform. Since 2022 I work as an independent principal engineer through Sarana Technology, delivering backend and AI platforms for product, AI, and public-sector clients, most recently leading a team of 7–10 engineers.

CORE SKILLS

- **Languages:** Go (primary), PHP/Laravel, Python, TypeScript/Node.js
- **Architecture:** Microservices, DDD, Clean Architecture, event-driven, CQRS, SAGA
- **Data & messaging:** PostgreSQL, MySQL, Redis, ScyllaDB, MongoDB · gRPC, Kafka, NATS, WebSocket (Centrifugo)
- **Infrastructure:** Kubernetes, Docker, AWS, Terraform, GitHub Actions
- **Observability:** Prometheus, Grafana, OpenTelemetry, ELK
- **Security:** OWASP Top 10, OAuth2, RBAC, SIEM/SOC, Trivy, Nuclei, Gitleaks, ZAP

PROFESSIONAL EXPERIENCE

Founder & Principal Engineer — Sarana Technology | 2022 – Present

Independent software consultancy. I work on contract as principal engineer or tech lead; each client engagement is listed below as its own role.

Technical Architect — Sonz.ai | Oct 2025 – Jun 2026

Contract via Sarana Technology. AI companion platform on Nakama and ScyllaDB with tiered LLM routing.

- Led a team of 7–10 engineers as de facto engineering manager: ran sprint planning, interviewed and hired engineers, held 1:1s and mentored, and set code review standards and the team's security and coding direction.
- Identified and fixed critical vulnerabilities in the production backend (auth bypass, CORS misconfiguration, leaked API keys) and reported the risks and fixes directly to the CEO.
- Designed and implemented the SAGA pattern for distributed in-app purchase and subscription workflows, ending recurring data-inconsistency incidents.
- Redesigned the AI memory system for reliability and observability and led its implementation, cutting debugging time for multi-turn conversation bugs by ~70%.
- Stayed on as the last engineer until the company wound down after it did not secure further funding.

Tech Lead — Pemerintah Kabupaten Muara Enim | Apr 2025 – Oct 2025

Contract via Sarana Technology. E-government and smart city platform for a regional government.

- Led a team of 3–4 engineers to design and deliver the platform backend, from architecture through production deployment.
- Deployed on Kubernetes with zero-downtime releases, horizontal autoscaling, and staging-to-production parity.
- Set the data security baseline to SPBE requirements: encryption at rest and in transit, RBAC, hardened API gateway, audit logging.
- Built an in-house mini SIEM that ingests logs, correlates security events, and alerts in real time, replacing a paid third-party SIEM.

Principal Engineer — Sangkuriang | Aug 2024 – Apr 2025

Contract via Sarana Technology.

- Migrated a monolith to Go microservices using DDD, speeding up feature delivery by 50% and reducing merge conflicts.
- Introduced gRPC for internal service calls, reducing end-to-end latency by 60% on critical user journeys.
- Drove TDD adoption to 90%+ test coverage, reducing production defects.

Backend Principal Engineer — Kipas Kipas | Aug 2023 – Mar 2024

Contract via Sarana Technology.

- Scaled the social platform backend to 500K+ active users at <200ms p95 response time.
- Doubled throughput with a Redis caching layer and async job processing, postponing an infrastructure upgrade by two quarters.

Technical Architect — Simajji SBH SDN | May 2021 – Aug 2023

- Designed architecture across multiple enterprise systems and led a microservices migration that improved performance by 40%.
- Standardized code review and release processes across distributed teams.

Backend Engineer — Rudex Tech (PT Rudex Teknologi Indonesia) | Nov 2018 – Feb 2021

Cybersecurity company specializing in SOC, penetration testing, and smart security products.

- Built PHP/Laravel backend services for SOC tooling, encrypted document management, and smart access and visitor management.
- Applied secure API design: authentication hardening, encrypted storage, input validation, safe handling of sensitive client data.
- Worked with security engineers and pentesters to fix findings and ship patches quickly.

Game Backend Developer — Sintask | 2014 – 2018 (part-time, while studying)

- Built server-side game logic in PHP and MySQL, with some JavaScript, for scoring, leveling and progression, and related player systems.
- Designed backend APIs and data models serving the game client.

SELECTED PROJECTS

- **emailblast** (github.com/bryanwahyu/emailblast) — High-throughput email campaign sender in Go. Benchmarked at 1 million emails in 21.1 seconds (~47K/s, 270 MB peak RSS, mock backend). Crash-safe: after a kill -9 at 446,574 sends, a rerun resumed from the checkpoint with no duplicates. Worker pool with rate limiting, SMTP/SES backends, dedup, dead-letter queue, dry-run mode, CI with race-detector tests.
- **Mini SIEM** (github.com/bryanwahyu/mini-siem-api) — Threat detection engine in Go that detects brute force, SQL injection, XSS, path traversal, scanners, and HTTP floods. Ingests from journald, file tailing, and HTTP; responds automatically via UFW, nftables, or Cloudflare; Prometheus metrics and offline-first storage to MinIO.
- **Automaton Sec** (github.com/bryanwahyu/automaton-sec) — DevSecOps platform in Go that runs Trivy, Nuclei, Gitleaks, OWASP ZAP, and SQLMap behind one REST API, with AI-assisted vulnerability analysis and real-time scan progress.
- **Disabilitasku.id & Sign Language AI** (github.com/bryanwahyu/app_disabilitas_isyarat_ai) — Founder of an accessibility platform for Indonesia. Built a computer-vision model that recognizes Indonesian sign language (BISINDO/SIBI) using MediaPipe and TensorFlow, packaged with Docker and GitHub Actions CI.

EDUCATION

Bachelor of Computer Science (S1 Teknik Informatika) — Udayana University · 2013 – 2018